BILL OF MATERIALS

28BYJ-48 Stepper Linear Actuator with Printed Ball Screw

Rev. 2026-10-20 13 items

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BOM	Product/Project: 28BYJ-48 Stepper Linear Actuator with Printed Ball Screw Revision date: 2026-10-20	https://www.printables.com/@TomKnox_1573/models

MOTION

1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	28BYJ-48 5V stepper motor with ULN2003 driver board	**1**	AliExpress Link	Amazon Link	*The upstream design is built around this motor. One motor + driver per slide.*

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CONTROLLER 1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Arduino Nano V3 ATmega328P CH340 with soldered headers	**1**	AliExpress Link	Amazon Link	*Runs the example sketch; an Uno works with the same pins.*

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SENSORS

1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	KW12-3 micro limit switch with roller lever	**1**	AliExpress Link	Amazon Link	*Endstop for homing (our addition; the upstream slide has no endstop). Mount at the motor end of the rail.*

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HARDWARE 4 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	4.5 mm steel BBs 0.177 cal 1000 pcs	**1**	AliExpress Link	Amazon Link	*Upstream: the linear bearing takes 34 + 6 BBs and the thrust bearing 8 (the short list says 8 + 8; pack per the assembly text). Steel, not*
>>	M3 x 10 mm hex socket cap screws 50 pcs	**1**	AliExpress Link	Amazon Link	*Upstream needs 12 x M3 SHCS 10 mm (end caps, flange, motor).*
>>	M3 x 20 mm hex socket cap screws 50 pcs	**1**	AliExpress Link	Amazon Link	*Upstream needs 2 x M3 SHCS 20 mm for the carriage preload adjustment.*
>>	M3 stainless hex nuts 100 pcs	**1**	AliExpress Link	Amazon Link	*Upstream needs 14 x M3 nuts.*

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POWER

2 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	5V 2A DC power supply 5.5x2.1mm barrel plug	**1**	AliExpress Link	Amazon Link	*Powers the ULN2003 board; do not run the motor from the Nano 5 V pin for long.*
>>	DC barrel jack 5.5x2.1mm female to screw terminal adapter	**1**	AliExpress Link	Amazon Link	*Supply to the driver board screw terminals.*

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WIRING 1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Dupont jumper wire set male-female male-male 40 pin	**1**	AliExpress Link	Amazon Link	*Driver IN1-IN4 and endstop wiring.*

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CONSUMABLES 2 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	PTFE lithium grease 50 g tube	**1**	AliExpress Link	Amazon Link	*Upstream recommends grease on the thrust bearing, drive screw and nut (not on the linear-bearing BBs).*
>>	PLA filament 1.75 mm 1 kg	**1**	AliExpress Link	Amazon Link	*All parts print in PLA. Consumable row; printed parts are not purchase rows.*

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WIRING 1 items

	Description	Qty	AliExpress	Amazon (Geniuslink)	Notes
>>	Mini USB cable 1 m data	**1**	AliExpress Link	Amazon Link	*Programming cable for the Nano (check whether your clone uses Mini, Micro or USB-C).*